FORMATION, COMPOSITION & NERVOUS REGULATION OF SALIVA & EFFECTS OF ALTERED SECRETION

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Lecturer

DIKIOHS, DUHS.

LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- ✔ Define saliva & describe its major functions in the oral cavity.
- ✔ Describe the process of saliva formation.
- ✔ Identify major & minor salivary glands.
- ✔ Describe the composition of saliva including water content, electrolytes, enzymes, mucins, antimicrobial substances & immunoglobulins.
- ✔ Explain the neural regulation of salivary secretion including parasympathetic control & sympathetic control.
- ✔ Define xerostomia & sialorrhea & their causes.
- ✔ Identify the effects of decreased salivary secretion.
- ✔ Correlate altered salivary secretion with common clinical conditions.

What is saliva?

Saliva is a watery liquid that the salivary glands secrete into the mouth

- Saliva is the first digestive juice present in the mouth which is secreted by salivary glands.
- The submandibular glands produce 70% of saliva, parotids produce 25% & sublingual glands contribute 5% of total salivary secretion.

FORMATION OF SALIVA

- Saliva formation occurs in two stages within the salivary glands.
- Acinar cells first produce isotonic primary saliva rich in water, NaCl & enzymes which is then modified into hypotonic saliva by ductal cells reabsorbing sodium & chloride while secreting potassium & bicarbonate.
- This fluid is then secreted mainly by major salivary glands under the control of autonomic nervous system.

STAGE 1: Secretion of primary saliva by acinar cells

Acinar cells produce primary saliva by actively transporting chloride into the lumen, followed by sodium & water.

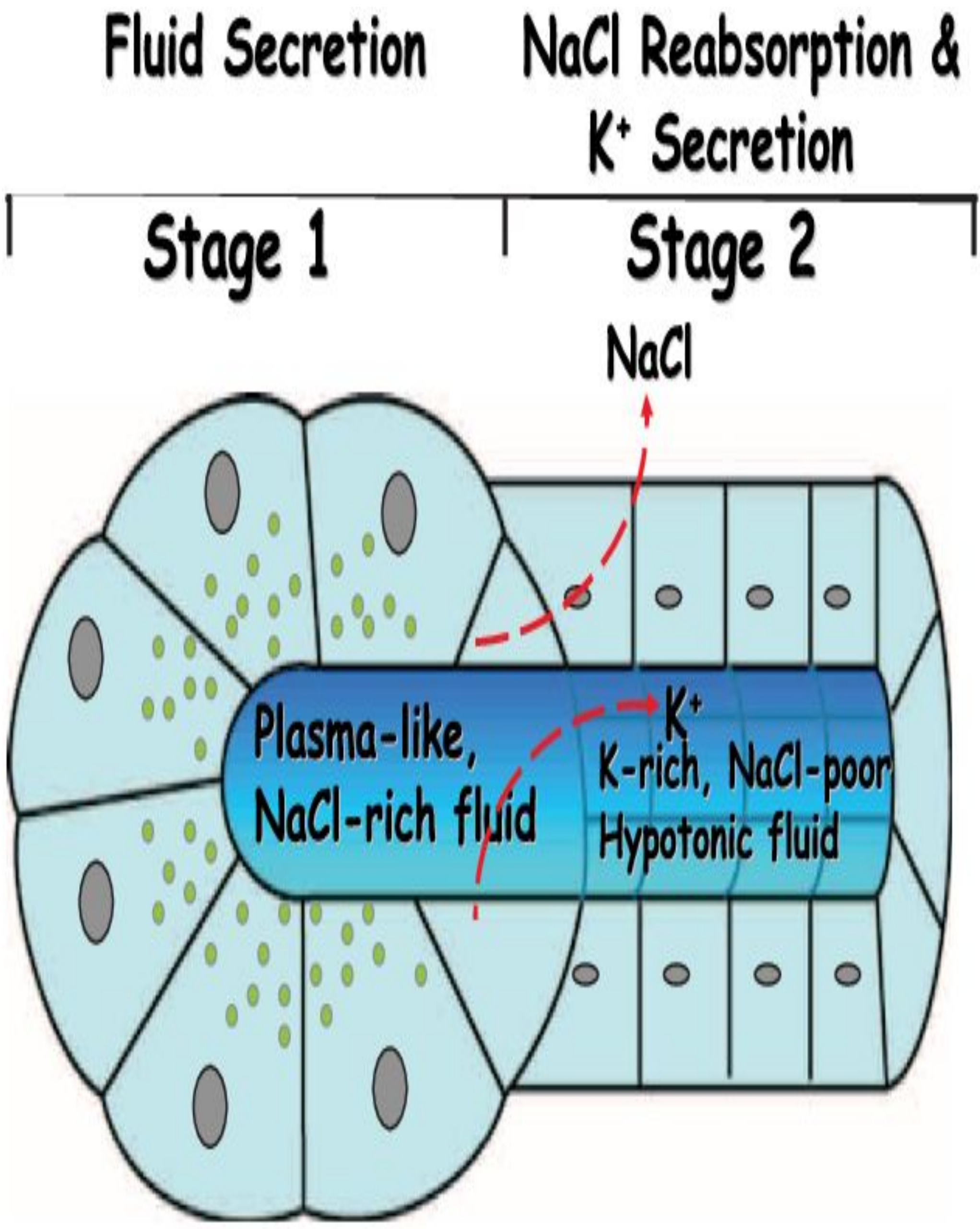
This saliva is ***isotonic*** compared to plasma.

CONT..

STAGE 2: Modification of primary saliva by ductal cells

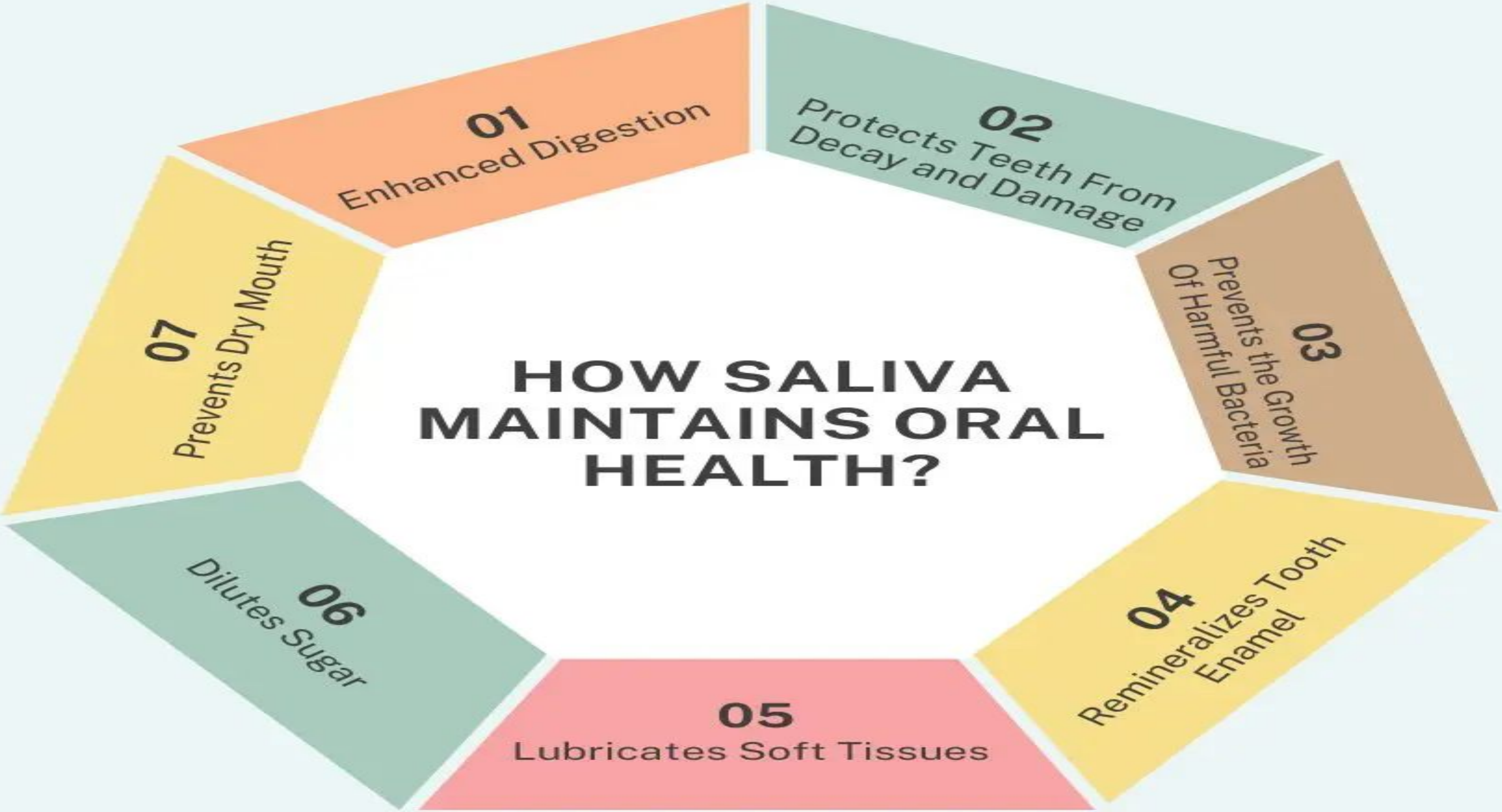
Primary saliva secreted by the acinar cells is then modified as it moves through the striated & excretory ducts.

Duct cells reabsorb Na^+ & Cl^- but have low water permeability which leads to a net reduction of solute, making the saliva ***hypotonic***(low salt conc.) before it enters the oral cavity.



FUNCTIONS OF SALIVA

- Lubricates the oral cavity.
- Moistens the food so it’s easier to chew & swallow.
- Starts the digestion process(***amylase*** in saliva helps breakdown starch).
- Protects against infection(***lysozyme*** disintegrates bacteria & prevent overgrowth of microbial population).
- Protects the teeth(***calcium hydroxyapatite*** prevents demineralization of teeth & washes away food debris from the teeth).
- Helps maintain pH balance in the mouth(maintain pH of 6.0 to 7.5 within the mouth).
- Helps us to taste food(dry food needs moisture for the taste buds to pick up the flavors).

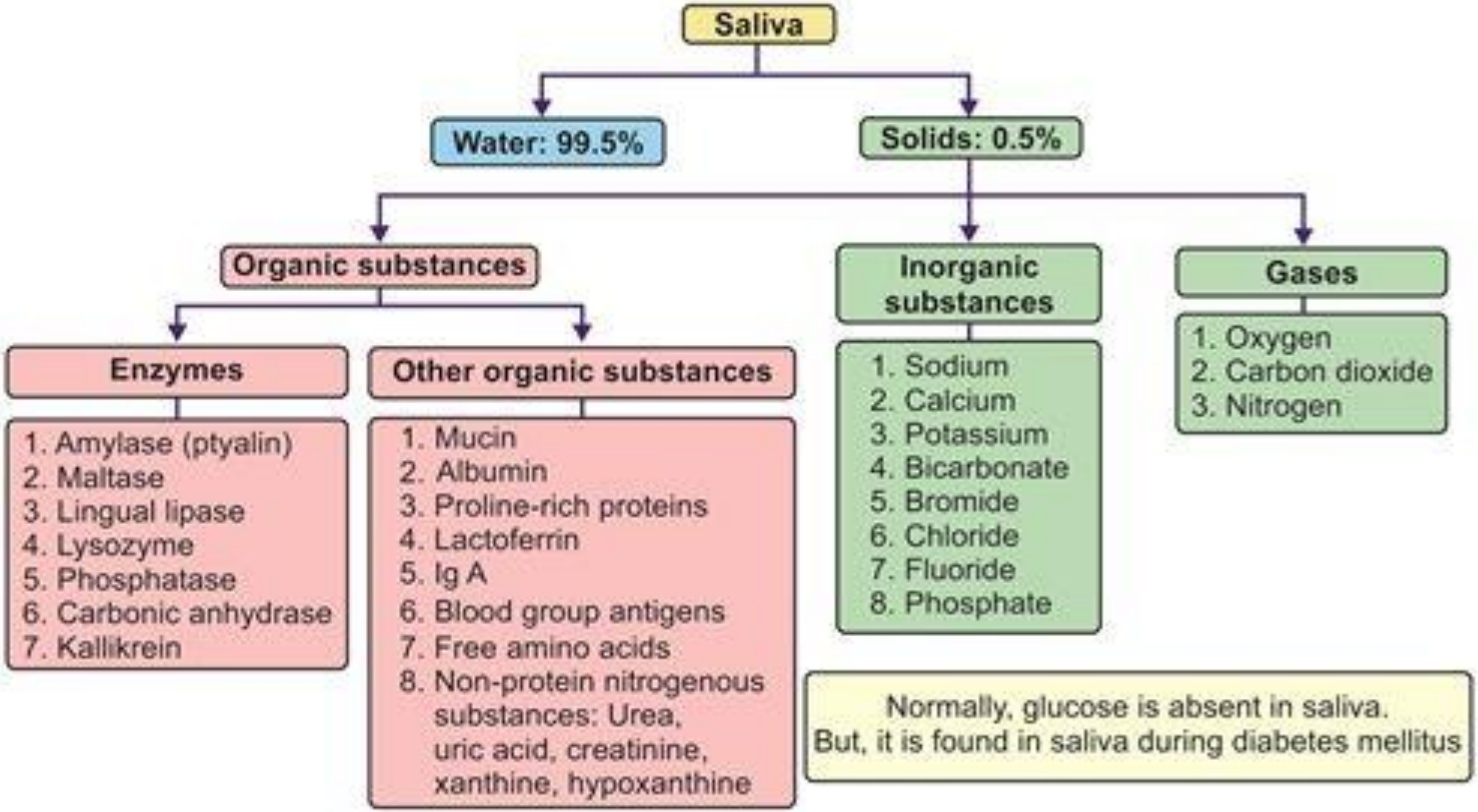

HOW SALIVA MAINTAINS ORAL HEALTH?
01
Enhanced Digestion
02
Protects Teeth From Decay and Damage
03
Prevents the Growth Of Harmful Bacteria
04
Remineralizes Tooth Enamel
05
Lubricates Soft Tissues
06
Dilutes Sugar
07
Prevents Dry Mouth

COMPOSITION OF SALIVA

- Saliva is a complex slightly acidic or neutral fluid composed of approximately 99% water.
- The remaining 1% consists of enzymes, electrolytes, antimicrobial compounds & mucin.

The major components of saliva are:

1. ***Water*** -> 99%
2. ***Electrolytes*** -> sodium(Na^+), potassium(K^+), calcium(Ca^{++}), chloride(Cl^-), bicarbonate(HCO_3^-) & phosphate(PO_4^{-3})
3. ***Enzymes*** -> amylase(ptyalin) & lingual lipase
4. ***Proteins & mucus*** -> mucin, immunoglobulin(IgA), lysozyme & peroxidase, statherin & proline rich proteins.
5. ***Trace amounts*** of urea, ammonia, glucose hormones & dissolved gases.


Saliva
Water: 99.5%
Solids: 0.5%
Organic substances
Inorganic substances
Gases
Enzymes
Other organic substances
1. Amylase (ptyalin)
2. Maltase
3. Lingual lipase
4. Lysozyme
5. Phosphatase
6. Carbonic anhydrase
7. Kallikrein
1. Mucin
2. Albumin
3. Proline-rich proteins
4. Lactoferrin
5. Ig A
6. Blood group antigens
7. Free amino acids
8. Non-protein nitrogenous substances: Urea, uric acid, creatinine, xanthine, hypoxanthine
1. Sodium
2. Calcium
3. Potassium
4. Bicarbonate
5. Bromide
6. Chloride
7. Fluoride
8. Phosphate
1. Oxygen
2. Carbon dioxide
3. Nitrogen
Normally, glucose is absent in saliva.
But, it is found in saliva during diabetes mellitus

MAJOR SALIVARY GLANDS

There are three major salivary glands i.e.

1. Parotid glands
2. Submaxillary or submandibular glands
3. Sublingual glands

PAROTID GLANDS

- The largest of all salivary glands.
- Situated at the side of face just below & in front of the ear.
- Secretions from these glands are emptied into the oral cavity by ***Stensen's duct***.
- Stensen's duct opens inside the cheek against the upper second molar tooth.

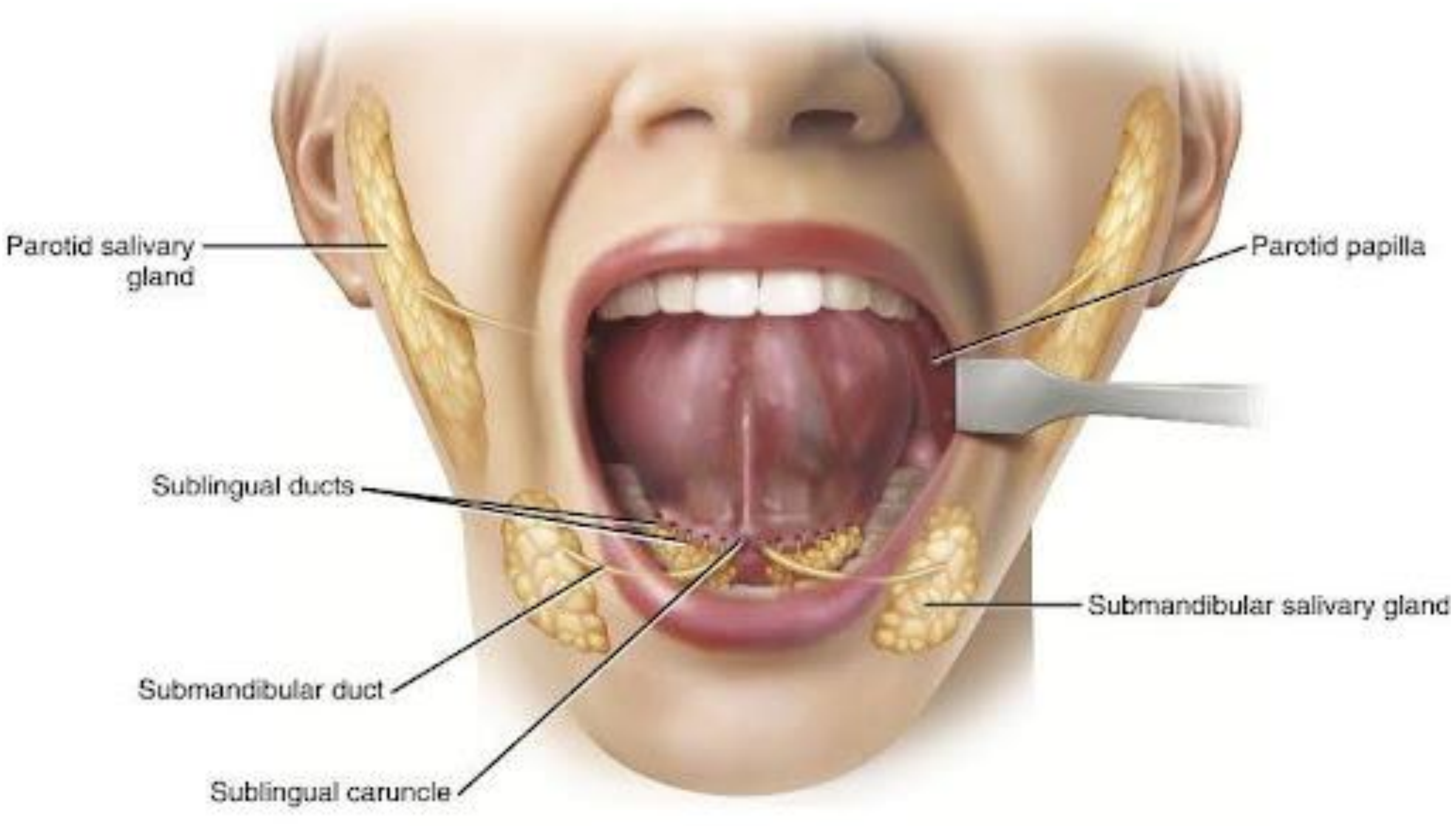
CONT..

SUBMANDIBULAR GLANDS

- Located in the submaxillary triangle medial to the mandible.
- Saliva from these glands is emptied into the oral cavity by ***Wharton's duct***.
- This duct opens at the side of frenulum of tongue by means of a small opening on the summit of papilla called ***caruncula sublingualis***.

SUBLINGUAL GLANDS

- These are the smallest salivary glands.
- Situated in the mucosa at the floor of the mouth.
- Saliva from these glands are poured into 5 to 15 small ducts called ***ducts of Rivinus***.



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MINOR SALIVARY GLANDS

BUCCAL GLANDS

- Buccal glands or molar glands are present between the mucous membrane & buccinator muscle.
- 4 to 5 of these glands are larger & situated outside the buccinator muscle around the terminal part of parotid duct

LABIAL GLANDS

- These glands are situated beneath the mucous membrane around the orifice of mouth.

PALATAL GLANDS

- Palatal glands are present beneath the mucous membrane of soft palate.

CONT..

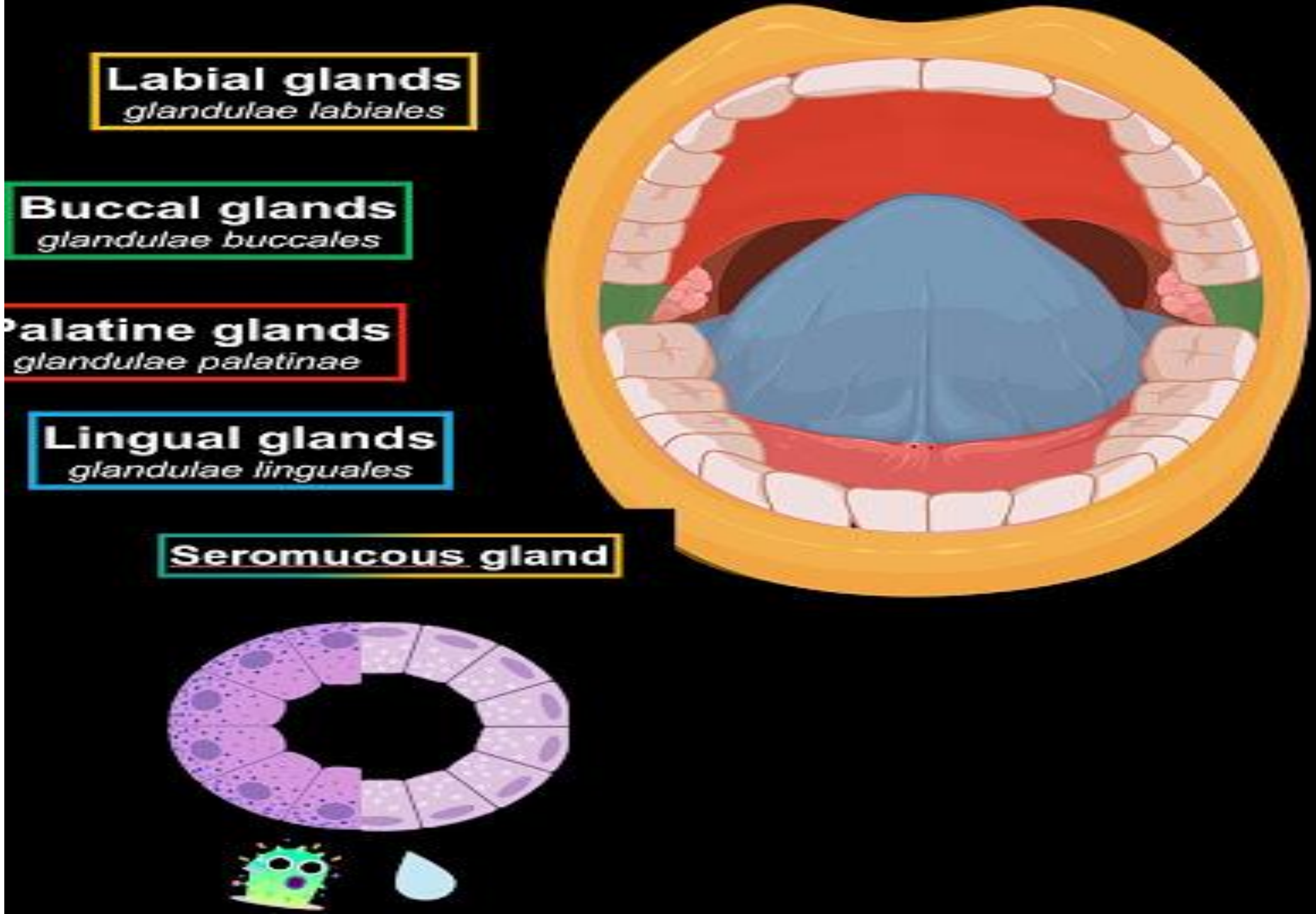
LINGUAL MUCOUS GLANDS

- These glands are situated in posterior one-third of the tongue behind ***circumvallate papillae*** & at the tip & margins of the tongue.

LINGUAL SEROUS GLANDS

- Lingual serous glands are located near ***circumvallate papillae*** & ***filiform papillae***.

Minor Salivary Glands [4]

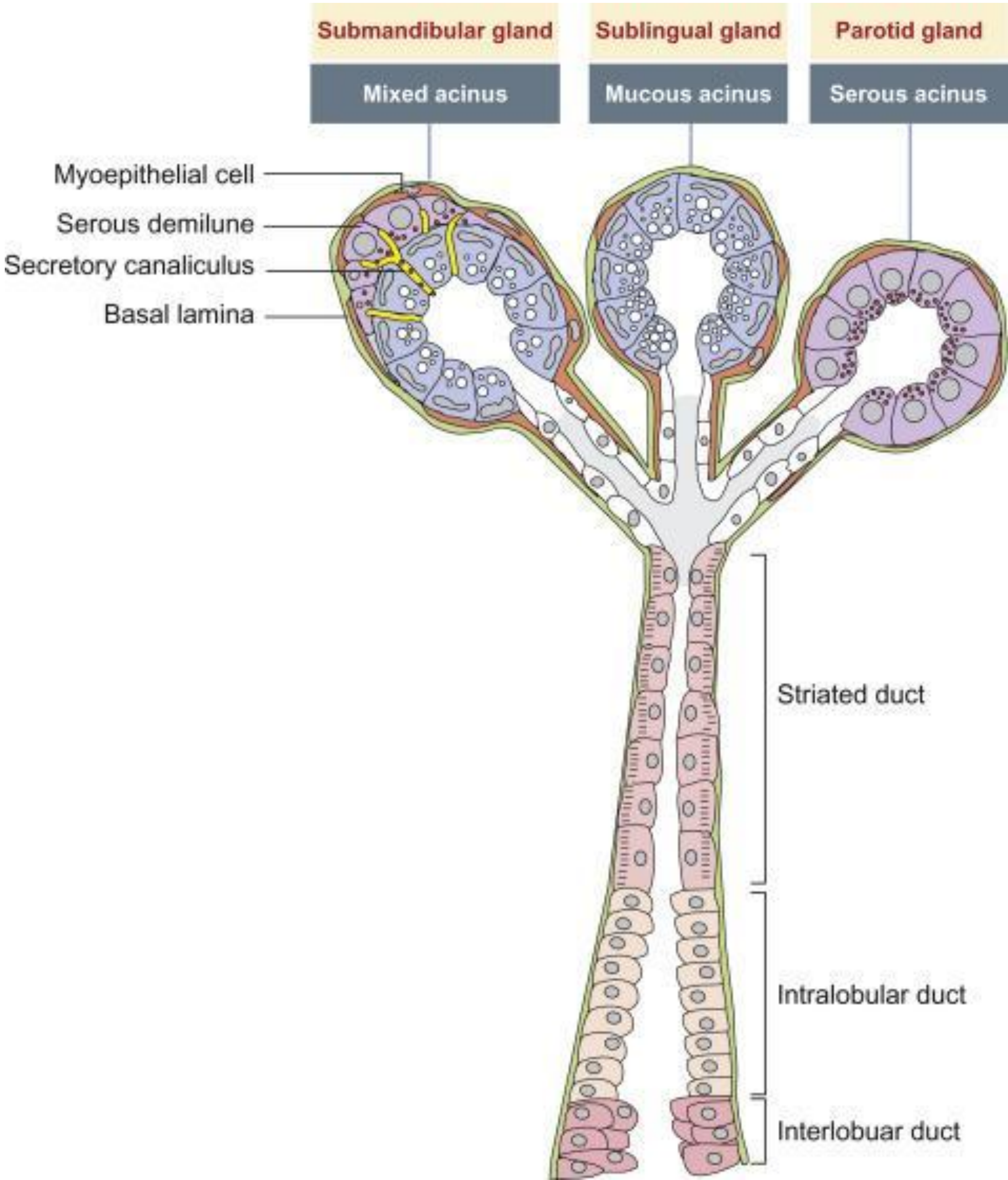


SEROUS & MUCOUS SALIVA

- Saliva contains two major types of protein secretion:

1. ***serous secretion*: ** that contains ptyalin(an alpha-amylase which is an enzyme for digesting starch).
2. ***mucous secretion*: ** that contains mucin for lubricating & surface protective purposes.

- The parotid glands secrete almost entirely the serous type of secretion & the sublingual glands secrete mucous secretions whereas submandibular glands secrete both serous & mucus secretions.



SEROUS	MUCOUS
Serous gland is a constituent of salivary glands, producing a solution with proteins in an isotonic watery fluid	Mucous gland is a constituent of salivary glands, producing a slippery, aqueous secretion
Produces a thin, watery secretion, comprising zymogens, antibodies, and inorganic ions	Produces a thick, viscous secretion, comprising of mucin
Cells contain round, central nuclei	Cells contain flattened nuclei against the basement membrane
Cells contain dispersed chromatin	Cells contain condensed chromatin
Stained in dark due to the presence of zymogens	Stained in light due to the presence of mucin
Contain large, striated ducts	Contain small, striated ducts
Involved in solubilizing dry food, maintaining oral hygiene, and initiating starch digestion	Involved in lubricating the oral cavity and making food into the slippery bolus

NERVOUS REGULATION OF SALIVARY SECRETION

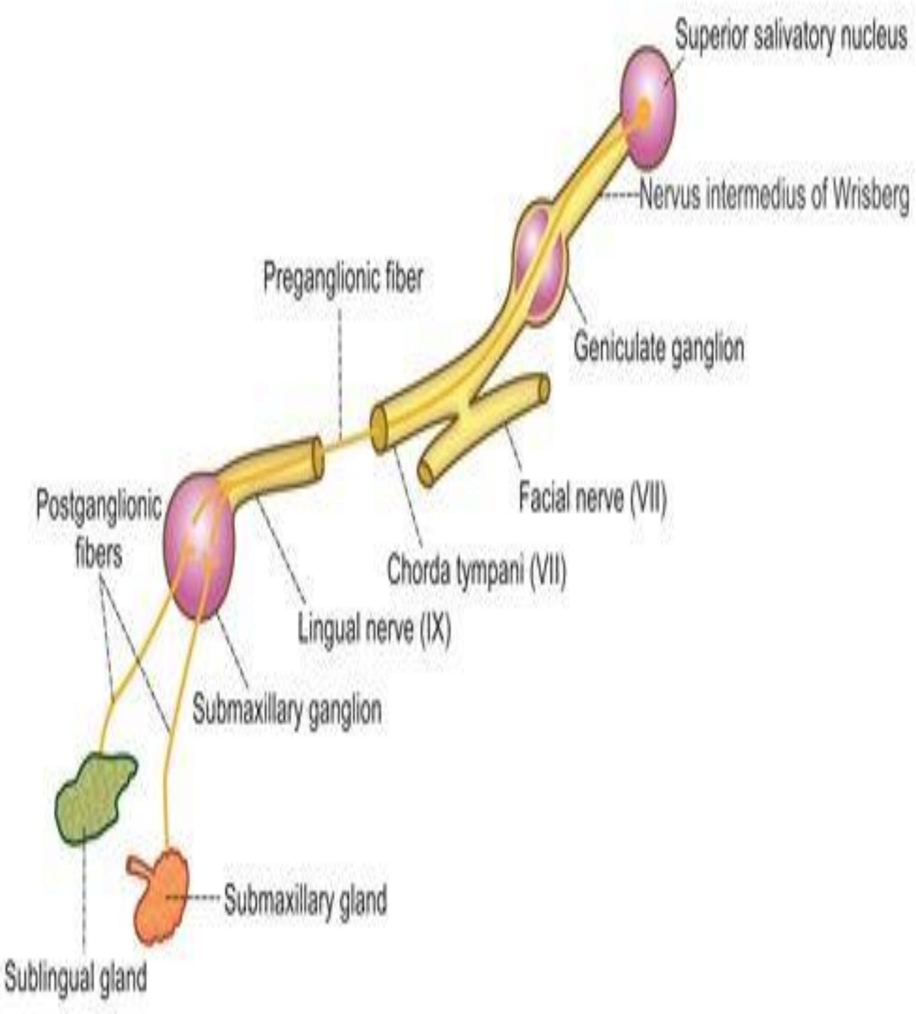
- Autonomic nervous system is involved in the regulation of salivary secretion.
- Although salivary glands are supplied by both parasympathetic & sympathetic divisions of autonomic nervous system, these glands are controlled mainly by parasympathetic nervous signals all the way from the superior & inferior salivatory nuclei in the brain stem.
- The salivatory nuclei are located approximately at the juncture of medulla & pons & are excited by both taste & tactile stimuli from the tongue & other areas of mouth & pharynx.
- Many taste stimuli, especially the sour taste can elicit copious salivary secretion(often 8 to 20 times the basal rate of secretion).

CONT..

- Certain tactile stimuli, such as presence of smooth objects in the mouth(e.g. a pebble) cause marked salivation, whereas rough objects cause less salivation & occasionally even inhibit salivation.
- Salivation can also be stimulated or inhibited by nervous signals arriving in the salivatory nuclei from higher centers of the central nervous system e.g. when a person smells or eats favourite foods, salivary secretion is greater than when food that is disliked is smelled or eaten.
- Salivation also occurs in response to reflexes originating in the stomach & upper small intestine, particularly when irritating foods are swallowed or when a person is nauseated.
- The saliva when swallowed, helps to remove the irritating factor by diluting or neutralizing the irritant substance.

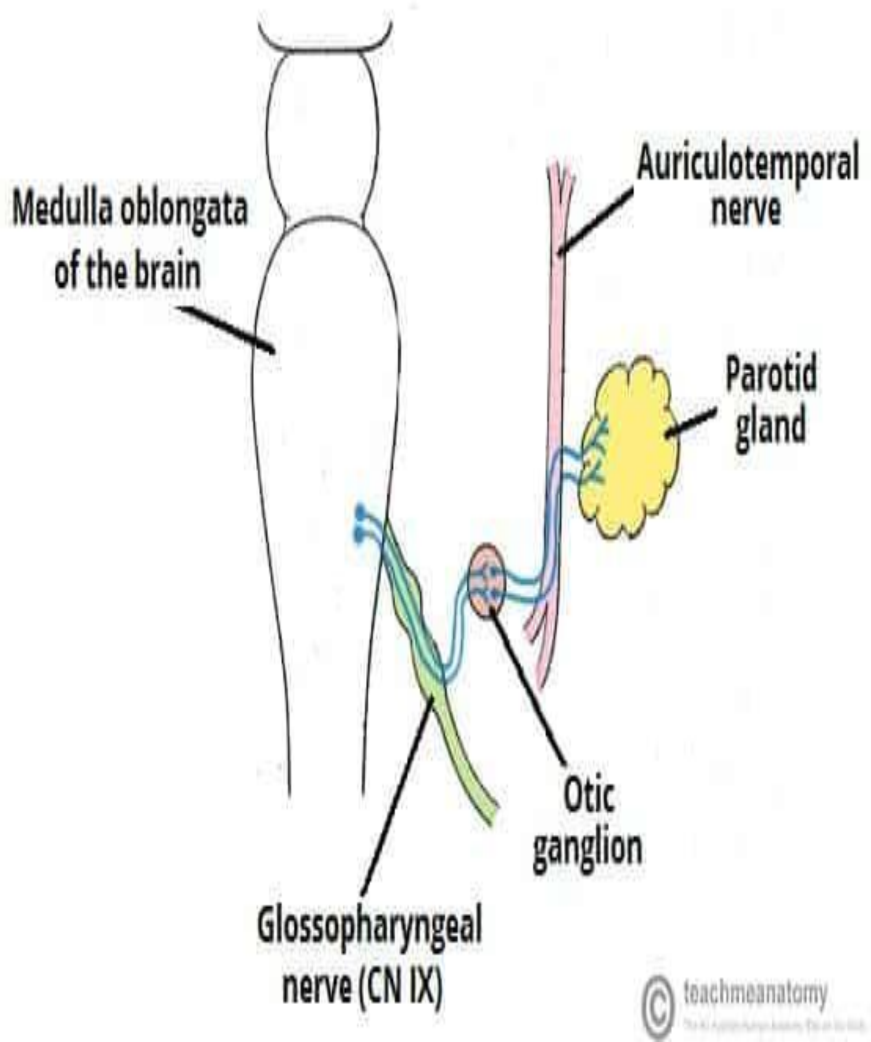
PARASYMPATHETIC SUPPLY TO SUBMANDIBULAR & SUBLINGUAL GLANDS

- Parasympathetic preganglionic fibers to submandibular & sublingual glands arise from the ***superior salivatory nucleus*** in the pons.
- After taking origin, the preganglionic fibers run through ***nervus intermedius of Wrisberg***, ***geniculate ganglion***, the motor fibers of ***facial nerve***, chorda tympani branch of facial nerve & lingual branch of ***trigeminal nerve*** & finally reach the ***submaxillary ganglion***.
- Postganglionic fibers arising from the submaxillary ganglion supply the submandibular & sublingual glands.



PARASYMPATHETIC SUPPLY TO PAROTID GLAND

- Parasympathetic preganglionic fibers to parotid gland arise from inferior salivatory nucleus situated in the upper part of medulla oblongata.
- From here, the preganglionic fibers run through the tympanic branch of ***glossopharyngeal nerve*** & ***tympanic plexus*** & end in ***otic ganglion***.
- Postganglionic fibers arising from the otic ganglion supply the parotid gland by passing through auriculotemporal branch in mandibular division of trigeminal nerve.



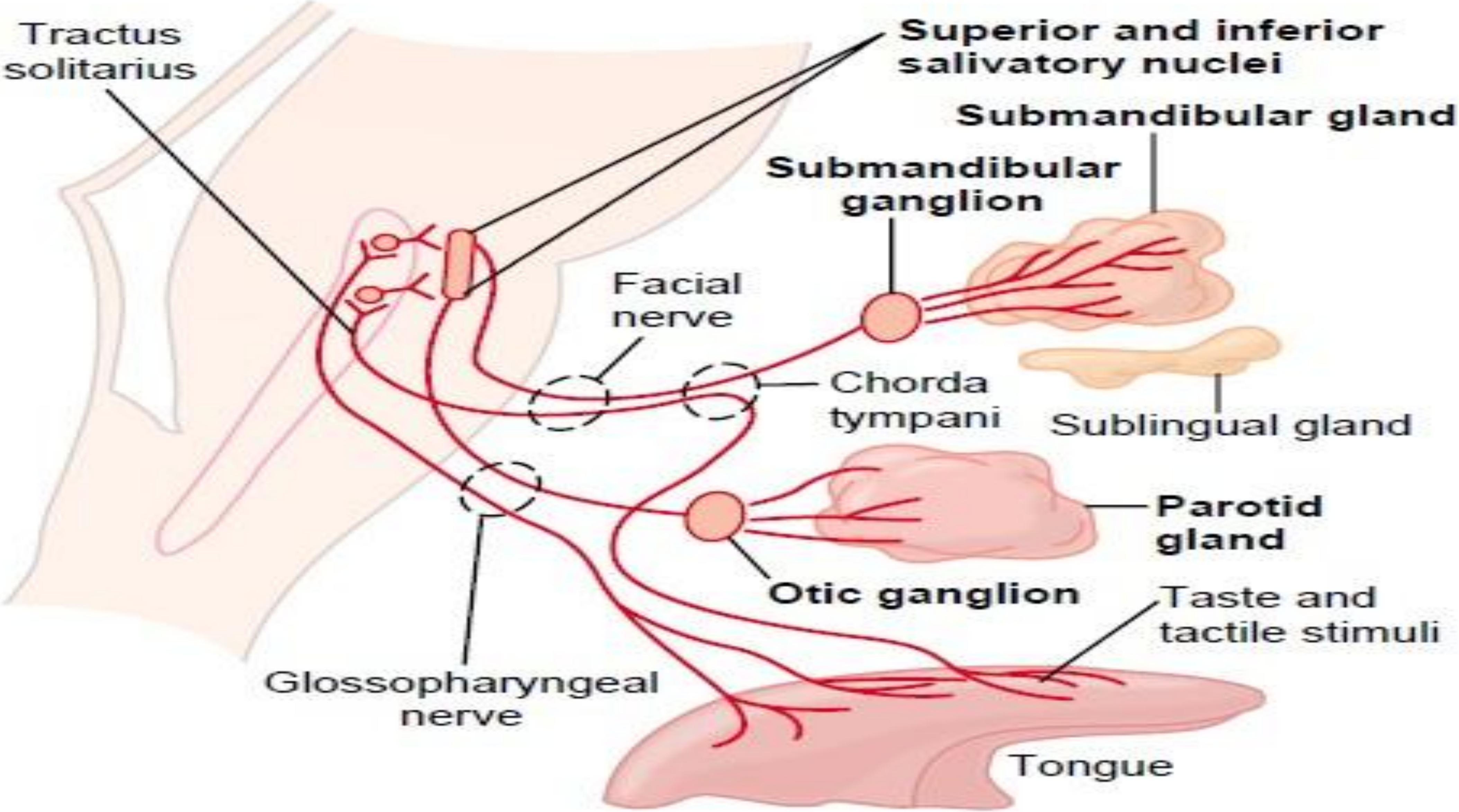


Figure 64–3

Parasympathetic nervous regulation of salivary secretion.

EFFECTS OF PARASYMPATHETIC STIMULATION

- Secretion of watery saliva.
- Vasodilation & activation of acinar cells.
- Less amount of organic constituents in saliva.
- Neurotransmitter -> acetylcholine

SYMPATHETIC SUPPLY TO SALIVARY GLANDS

- Sympathetic preganglionic fibers to salivary glands arise from the lateral horns of first & second thoracic segment of spinal cord.
- These fibers leave the cord through the anterior nerve roots & end in ***superior cervical ganglion*** of the sympathetic chain.
- Postganglionic fibers arising from the ganglion supply the salivary glands along the nerve plexus around the arteries supplying the glands.

EFFECTS OF SYMPATHETIC STIMULATION

- Thick saliva rich in organic constituents like mucus.
- Vasoconstriction & activation of acinar cells.
- Neurotransmitter -> noradrenaline

REFLEX REGULATION OF SALIVARY SECRETION

 There are two types of salivary reflexes:

1. Unconditioned reflex
2. Conditioned reflex

UNCONDITIONED REFLEX: inborn reflex that is present since birth induces salivary secretion when any substance is placed in the mouth due to the stimulation of nerve endings in the mucous membrane of oral cavity.

CONDITIONED REFLEX: acquired reflex which needs previous experience. Presence of food in the mouth is not necessary to elicit this reflex & can be stimulated by sight, smell, hearing or even thought of food.

APPLIED PHYSIOLOGY

1. HYPOSALIVATION

Reduction in salivary secretion is called hyposalivation.

I. Temporary hyposalivation

II. Permanent hyposalivation

Temporary hyposalivation: occurs in emotional conditions(e.g. fear), fever, dehydration.

Permanent hyposalivation: occurs in sialolithiasis(obstruction of salivary duct), congenital absence or hypoplasia of salivary glands, Bell's palsy(paralysis of facial nerve).

2. HYPERSALIVATION

- Excessive secretion of saliva is called hypersalivation.
- Pregnancy is a physiological condition associated with hypersalivation.
- Hypersalivation associated with pathological condition(e.g. tooth decay, neoplasm, neurological disorders, nausea, vomiting) is called ptyalism, sialorrhea, sialism(sialosis).

CONDITIONS CAUSING ALTERED SALIVARY SECRETION

1. *CHORDA TYMPANI SYNDROME*

- It is a condition characterized by ***sweating while eating***.
- During trauma or surgical procedure, some parasympathetic nerve fibers to salivary glands may be damaged.
- During the regeneration, some of these nerve fibers which run along with chorda tympani branch of facial nerve may deviate & join with the nerve fibers supplying the sweat glands.
- When the food is placed in the mouth, salivary secretion is associated with sweat secretion.

2. PARALYTIC SECRETION OF SALIVA

- The increased secretion of saliva after cutting the parasympathetic nerve fibers is called paralytic secretion.
- When the parasympathetic nerve to salivary gland is cut in experimental animals, salivary secretion increases for the first three weeks & later it diminishes & finally stops at about sixth week.
- It is because of hyperactivity of sympathetic nerve fibers that supply the salivary glands after cutting the parasympathetic fibers.

3. SJOGREN'S SYNDROME

- It is an autoimmune disorder in which the immune cells destroy the exocrine glands(e.g. lacrimal glands, salivary glands).
- Common symptoms of Sjogren's syndrome are dryness of mouth due to lack of saliva(xerostomia), persistant cough & dryness of eyes.
- In some cases it may cause dryness of skin, nose & vagina.

4. MUMPS

- It is an acute viral infection of the parotid glands.
- The virus causing this disease is *paramyxovirus*.
- Clinical features of mumps are puffiness of cheeks due to swelling of parotid glands, fever, sore throat & weakness.

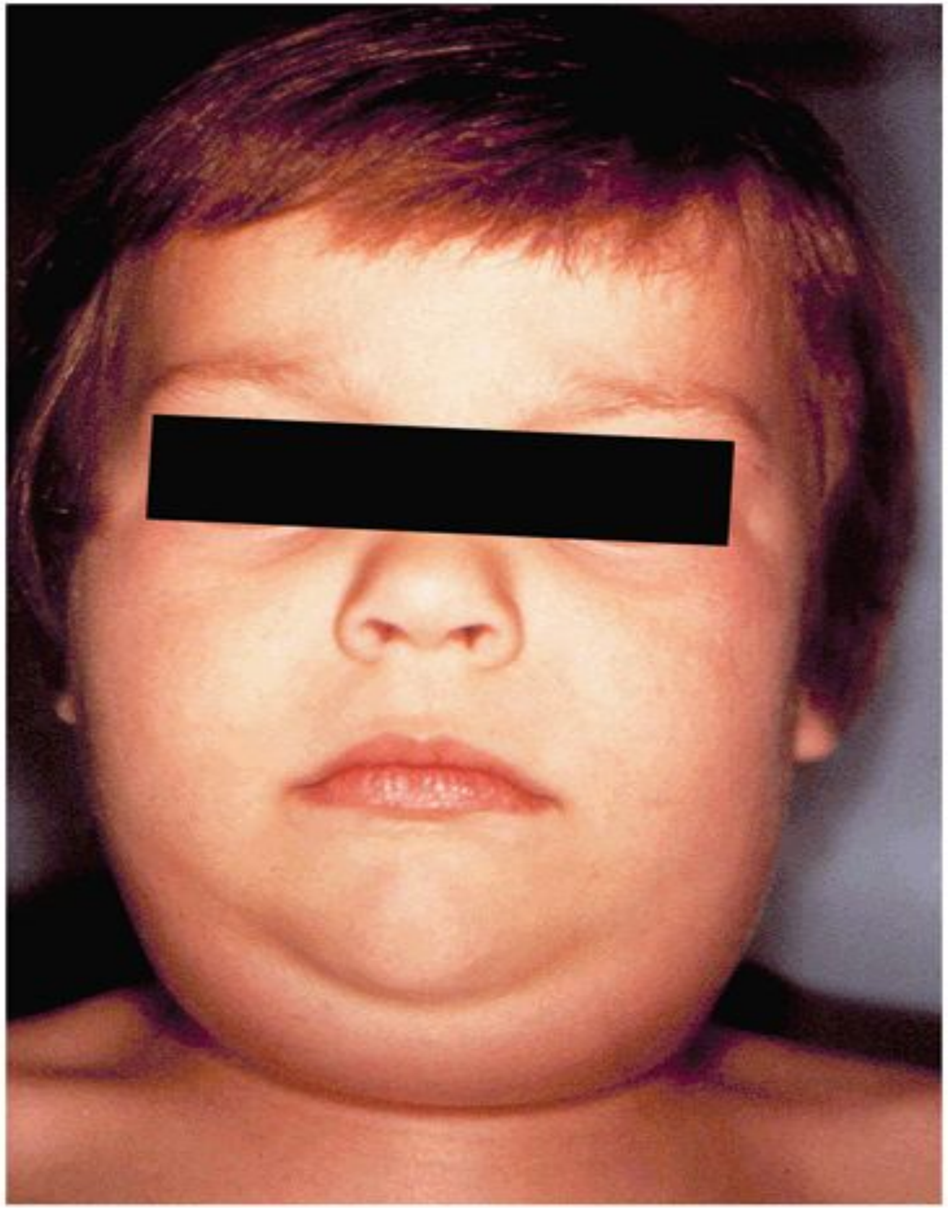


FIGURE 16.1 Child with mumps demonstrating neck swelling due to salivary gland enlargement. (Provided by the Centes for Disease Control and Prevention [CDC] Public Health Image Library, Patricia Smith and Barbara Rice, image #1861.)

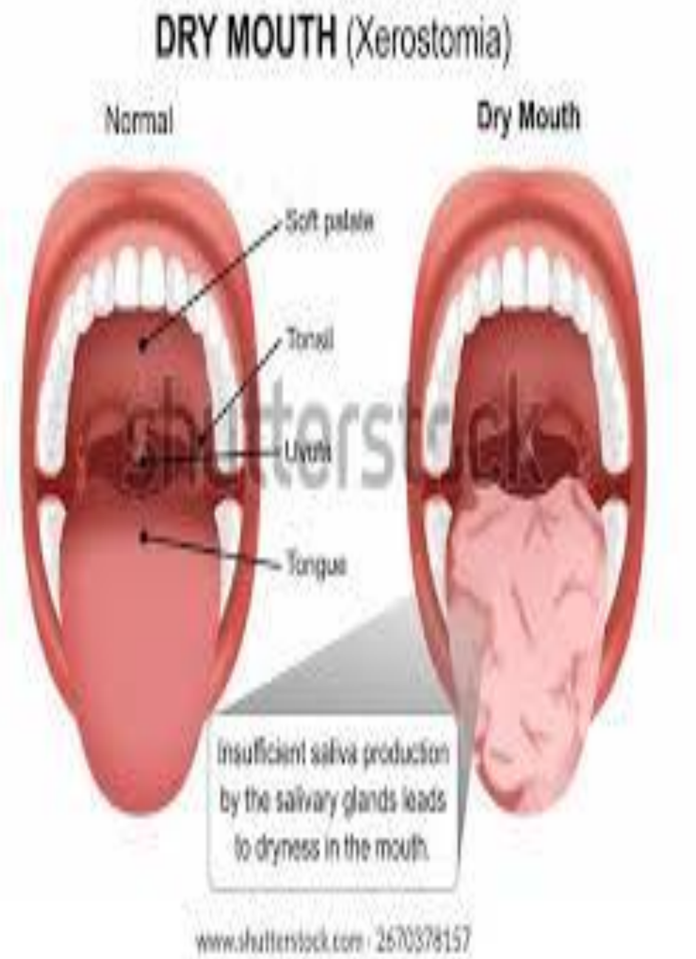
AUGMENTED SECRETION OF SALIVA

- If the nerves supplying the salivary glands are stimulated twice, the amount of saliva secreted by the second stimulus is more than the amount secreted by the first stimulus.
- It is because, the first stimulus increases the excitability of acinar cells so that when the second stimulus is applied, the salivary secretion is augmented.

EFFECTS OF ALTERED SECRETION

XEROSTOMIA

- Xerostomia means dry mouth.
- Due to hyposalivation or absence of salivary secretion.
- Common causes include dehydration or renal failure, Sjogren’s syndrome, radiotherapy, trauma to salivary glands or their ducts.
- Side effects of some drugs like antihistamines, antidepressants, antiparkinsonian drugs & antimuscarinic drugs can also cause xerostomia.
- Xerostomia causes difficulties in mastication, swallowing & speech.
- It also causes *halitosis*(bad breath)



DROOLING

- Inability to retain saliva within the mouth is called drooling.
- It occurs due to excessive production of saliva.
- Conditions that cause drooling include eruption of teeth in children, upper respiratory tract infections or nasal allergies in children, tonsillitis, difficulty in swallowing & peritonsillar abscess.

POST ASSESSMENT

Q1. Saliva becomes hypotonic mainly because:

a) Water is actively secreted into ducts

b) Sodium & chloride are reabsorbed in ducts

c) Proteins are removed from saliva

d) Potassium is completely reabsorbed

Q2. Which enzyme in saliva begins carbohydrate digestion:

a) Pepsin

b) Trypsin

c) Salivary amylase

d) Lipase

CONT..

Q3. Which condition is commonly associated with decreased salivary secretion:

a) Hyperthyroidism
b) Sjogren’s syndrome
c) Peptic ulcers
d) Diabetes insipidus

Q4. Secretion of saliva is regulated by:

a) Endocrine control only
b) Local tissue hormones only
c) Renal mechanism
d) Nervous reflexes


THANK YOU!